

Name: _____

Class: _____



Mr. Rawson • GCSE Computer Science

Year 10 2026

Binary and Denary Conversion

*Due Tuesday 8 September, start of the lesson. Show your working where a question asks for it. Every binary number on this sheet is **8 bits** — write all eight digits, including leading zeros. **Total / 17***

1) **Convert** each 8-bit binary number into denary.

[4]

8-bit binary	Denary
00001101	
00110010	
10000001	
11101100	

2) **Convert** each denary number into 8-bit binary.

[4]

Denary	8-bit binary
6	
40	
100	
173	

3) **Convert** the denary number 221 into 8-bit binary. Show your working.

[2]

4) **Complete** the table by writing in the missing value in each row.

[3]

Denary	8-bit binary
19	
	01100000
	11111111

5) **State** the range of denary numbers that can be represented by an 8-bit binary number.

[2]

(a) The **largest** denary number: _____

[1]

(b) The **smallest** denary number: _____

[1]

6) **Describe** how to convert an 8-bit binary number into denary. Use an example in your answer.

[2]
